

lyra-cpp — Session Log

Running EOD log. Newest entry on top. Short rough-outline format.

2026-06-25 — v0.4.10 RELEASED: CTUN drag-the-marker (4th look, FIXED) + CW decoder A-series + CW-decoder docs

Working on main. Two releases since v0.4.8: **v0.4.9** (fast patch — floating tool panels are float-only, Lock no longer disables the CW console/decoder; 162906f) and today's **v0.4.10** (ebb3c67).

CTUN #174 — 4th look, the pass that actually fixed it (PanadapterPanel.qml)

- Operator (Thetis Help open as ground truth) said CTUN still wasn't Thetis

after the v0.4.4/v0.4.5 "parity" rebuilds: you should drag the FREQ MARKER onto signals with the display LOCKED; a click INSIDE the filter must NOT jump the 0-beat.

- Read the real Thetis mouse handlers FIRST (console.cs 50230-50433 press +

51887-51960 drag). The DSP core (locked-LO + WDSP NCO shift) was already right; the whole gap was the MOUSE + MARKER layer.

- ROOT CAUSE the 3 prior passes all missed: Lyra's carrier marker is

hard-pinned to the locked display centre (markerOffsetHz = CW pitch only), so under CTUN there was NO marker tracking the dial — nothing to drag. First bench report: "marker does not move."

- Fix (CTUN-gated, non-CTUN byte-identical): new ctunDialOffsetHz slides the

carrier marker + filter passband to (rx1FreqHz – centre); body-drag = relative 1:1 marker-grab (setRx1FreqHz); top-15px strip = LO pan (setCtuneCenterHz); inside-passband click suppresses the retune; the edge-drag + inPassband math subtract the dial offset.

- Operator-bench-CONFIRMED "Do as Thetis does state." #174 DONE.

CW decoder — rest of the safe Tier-A A-series (A4/A6/A7/A8) (CwDecoder.cpp/.h)

- A-slice A1/A2/A3/A5 was committed last session (eb4c89d). v0.4.10 adds:

- A4 sub-bin AFC: 3-pt parabolic interp on the winning Goertzel bin

(refines bestHz only).

- A6 hysteretic SNR gate: open at squelch×floor, hold to 0.8× that

(≈1.2× floor); inert on clean signal.

- A7 unify the two word-gap emitters: guard the processEdge Word emit with

lastEmitted_ != ' ' so a word gap can't double-emit.

- A8 add _ / \$ punctuation; prosigns deliberately omitted (alias +/().

- All additive — correct decodes unchanged, ~35 tuned constants untouched.

Unit test ALL PASS (VVV TEST CQ / CQ PARIS / CQ TEST; AFC lock 749).

CW decoder docs (USER_GUIDE.md)

- Operator couldn't recall what the decoder NB / DSP-filter did → realised

none of the decoder was documented (+ a stale "decoder coming in a later release" line). Added "Reading CW — the RX decoder" section + TOC entry: panel, confidence dimming, decoded-call/name grab → His Call / Name (feeds {CALL}/{NAME}), Match-TX-to-RX-WPM, and every detector knob. Console His-call cross-linked to the decoder grab.

B1 anchor (NEXT CW work — NOT started)

- Benchd a live **W1AW** run; it announces its own speed ("END OF 30WPM

TEXT", "NOW 25 WPM") = labeled truth. Sent 30→25 wpm, decoder read **35** → over-reads ~15-20%, dit estimate runs short. Copy still readable; residual errors (WILL→WIRL, ONCE→JNCE) are valid-letter swaps A1 can't touch by design. W1AW = the labeled

bench to tune B1 (independent dah-length + warm-up/Gaussian boundary fix).

Release mechanics + standing

- 0.4.9 → 0.4.10 (CMakeLists / lyra.iss / README); notes

docs/releases/v0.4.10.md. Commit ebb3c67, tag v0.4.10, pushed main + tag, GitHub release + dist/Lyra-Setup-0.4.10.exe published.

- Daily backup task ("Lyra-cpp Daily Backup", 22:30 → _backups/ + mirror Z:\LyraBU\ confirmed ACTIVE + healthy (last run 6/24 OK, 0x0).
- Open: CW B1 (with the W1AW bench). CTUN done. Bigger arcs still open — RX2 #96-101, CW macro chips #176, grab-spot-as-call #182, PureSignal.

2026-06-24 — v0.4.8 RELEASED: panel docking overhaul + CW transmit metering (+ wiki, CW-decoder review)

Working on main (last tag v0.4.7 = ff6c9c9). Tester feedback (Randy W7CPA / Jim / a 3rd) said the main panels "all float, nothing docks, drop-against-an-edge does nothing," and resizing was a fight. Big UI arc + a CW metering arc, shipped as **v0.4.8**.

Panel docking overhaul (#184) — dockdragcontroller.{h,cpp} (NEW) + mainwindow.{h,cpp}

- Root cause: the custom dock title bar (makeDockTitleBar) replaces QDockWidget's built-in drag, killing Qt's drag->drop-indicator->snap-to-edge.
 - New DockDragController (public QMainWindow API only): press->threshold->drag->drop FSM on each title bar, translucent **cyan drop-zone overlay** — snap-to-edge (addDockWidget), neighbor split L/R/T/B thirds (splitDockWidget, "before"=double-split), center tabify (tabifyDockWidget), off-region float (setFloating). Save-on-commit.
 - Resize: dock separators widened 6px + cyan-on-hover; tab bar styled.
- Compact custom move cursor (arrow while locked). One-time status-bar hint.
- **Lock = move + resize**: disables move/float/close via features AND blocks the dock-separator resize press in event() (split-cursor detect). The earlier setFixedSize-on-lock froze panels and collapsed the layout on a locked restart — REMOVED; event-interception locks resize cleanly and always restores. (Two intermediate restart-collapse symptoms chased + fixed.)
 - View -> Layouts: **4 named slots + Lyra default** (recall no longer force-shows the chip-summoned TX/RX/CW rack panels).

Factory default fixed — default_layout.h + applyDefaultLayout

- Prior embedded default was base64 of the raw registry REG_BINARY -> carried the QSettings @ByteArray(. . .) UTF-16 wrapper -> restoreState() rejected it (the "screwed up default"). Re-embedded from operator's **N8SDR** slot, de-wrapped to the real saveState bytes, + defaultPanadapterSplit().

CW transmit metering (#105) — hl2_stream.{h,cpp}, metermodel.{h,cpp}, main.cpp, QML

- Bug: QSK CW keys the PA at the gateway and deliberately leaves wire MOX low, so the meter (gated on moxActive) never flipped to TX during CW (external watt-meter showed power). Other modes flip because they assert MOX.
- cwKeyingActive (message-level, from the CwKeyer onState hook, marshalled to the stream thread) drives the meter RX<->TX swap + the Ladder source set. Reads fwdPowerW() (already valid during gateway CW). Does NOT assert wire MOX — QSK keying unchanged.
- txDisplayActive (= moxActive || cwKeyingActive) drives the **red-on-air** QML (VFO border + RX/TX tag in TuningPanel, split/XIT TX markers in PanadapterPanel).

- **Paddle/key** detection: onHwPttPoll (~50 ms) latches cwKeyingActive

from **forward power** (≥ 0.5 W, ~500 ms hang), gated to CW mode, skipped while CWX drives it — this HL2+ reads non-zero ptt_in at rest (\$10 Q#1), so forward power is the clean signal.

- The panadapter spectrum **stays RX** during CW (gateway CW never feeds the

WDSP TX analyzer -> swapping shows an empty pan; RX shows the keyed carrier). Operator-confirmed regression + fix.

Release / docs / wiki

- Version 0.4.7->0.4.8 (CMake project(VERSION) + installer/lyra.iss).

docs/releases/v0.4.8.md; USER_GUIDE "Getting around the window" rewritten + Meter CW note; README Features->v0.4.8. ISCC -> dist/Lyra-Setup-0.4.8.exe (69 MB). Tag v0.4.8, pushed main+tag, GitHub Release published w/ installer.

- **GitHub Wiki initialized + built:** Home, User Guide (mirror of

USER_GUIDE.md), Installing-and-first-connection, FAQ-and-Troubleshooting, _Sidebar. Sources version-controlled in docs/wiki/ (User Guide generated from USER_GUIDE.md at publish). NB: the first wiki page must be created once in the web UI (no API, no push-init) — operator did it; rest pushed via ...wiki.git.

CW decoder accuracy review (analysis only — NOTHING changed)

- Recovered the 4-reviewer + code-cross-checked review (was transcript-only)

and committed it: docs/CW_DECODER_REVIEW.md (Tiers A–D). Target = src/dsp/CwDecoder.cpp/.h (field-tuned constants).

- Operator chose the safe **A-slice = A1+A2+A3+A5** to do next (nearest-code

? rescue / AFC center-bias / per-char confidence / crash-proof peak-snap — none touch the tuned constants; rule: never corrupt a correct decode). Then **B1** (independent dah-length + the $\sqrt{3}$ -vs-1.565 boundary discontinuity). NOT STARTED. See [[project-lyra-cpp-cw-decoder-accuracy]].

2026-06-23 — v0.4.6 RELEASED: connectivity + multi-radio + front-panel polish

Working directly on main (last release v0.4.5 4a106d5). Connectivity/ polish release — no wire/TX/audio-path changes; all UI / discovery / installer.

Installer (lyra.iss)

- Auto Windows Firewall **inbound** rules for lyra.exe: UDP (HL2 discovery

reply + EP6 RX stream) + TCP (TCI server, future networked clients). Named "Lyra SDR", delete-then-add (idempotent), removed on uninstall. **Fixes the "needs admin to connect" report** — the unsolicited inbound discovery reply was being dropped for a non-elevated launch. Matches Thetis's WiX FirewallException approach.

- Opt-in NetworkThrottlingIndex=0xffffffff task (default UNTicked) — the

classic multimedia network-throttle "gaming" tweak; [Registry], left in place on uninstall. Mirrors Thetis NetworkThrottle.cs.

Discovery (hl2_discovery.{h,cpp})

- **Dual broadcast:** sendBroadcastFromAllSockets now sends to both

255.255.255.255 AND each NIC's subnet-directed broadcast (entry.broadcast()).

- probe(ip) — directed unicast discovery for Add-by-IP (own socket +

1 s deadline, reuses parseReply→radioFound). Independent of the sweep.

Settings → Hardware radio panel (settingsdialog.cpp)

- **Add by IP** fires probe() → the placeholder upgrades in place to the real

board/gw/rx (addRadio changed skip-on-dup → update-in-place).

- **Double-click a row → Open; connected radio shown green + bold**

(item roles, survives re-render); one-line helper.

Front panel (mainwindow.{h,cpp})

- **Start green / Stop red** (backing QToolButton), ****Connected green / Disconnected red / Connecting-Scanning amber**** (connStatus_).
- Header chip strip: new **Options:** separator+label so **CTUN / WF-ID** no longer read as RX-DSP functions.

Release

- Version bump CMakeLists 0.4.5→0.4.6 + lyra.iss AppVersion. Release note docs/releases/v0.4.6.md; USER_GUIDE header + Radio sections; README Features (v0.4.6). Tag v0.4.6, installer rebuilt, pushed to origin/main + GitHub release.

2026-06-22 — v0.4.5 RELEASED: CTUNE rebuild + Waterfall ID + CW macros

Branch feature/ctune-rebuild-wfid; main was at v0.4.4. This is the v0.4.5 release.

#174 CTUNE rebuilt (operator: v0.4.4 "built wrong")

- 4-agent Thetis-vs-Lyra audit. Centre/shift/edge decision relocated from QML into the C++ RX-NCO chokepoint (HL2Stream::pushEffectiveRxFreq) so EVERY tune source behaves (dial / click / keypad / band+memory / TCI spot): smooth- scroll at the 5% display margin, far-jump re-centre, zoom force-centre, IQ- Nyquist clamp. RXANBPSetShiftFrequency added so notches/NB track the shift.
- Commit 29acc6d (design docs 0e0f2f2).

#175 Waterfall callsign ID

- TX courtesy ID: the call rendered as a readable image in the SSB passband. Header arm chip (USB/LSB only), Settings → TX config (level ≤0.065 / repeat / over-drive warning), crash-safe transient flat-TX orchestrator (TCI source + whole-rack bypass + DIGU/DIGL, restore on un-key).
- **LSB mirror bug fixed** this session: DIGL flips audio↔RF, so the call came out mirrored in LSB (USB was fine). Renderer now pre-mirrors the frequency layout for LSB → reads upright on both sidebands. Commit 79d8d43.

#176 CW macros — named click / F-key memories

- Redesigned floating CW console into a contest-grade macro bank: named click + F-key (F1–F12, global) memories, a "My macros" group the operator builds out, live "sending" state, Repeat, a contact row (His call / Name / RST / #).
- Tokens: contact {CALL}/{NAME}/{RST}/{#} + {MYCALL}, plus operator personal tokens ({ME}/{PWR}/{QTH}/{RIG}/...), with a click-to-insert palette.
- New CwMacroModel persists macros / tokens / serial to QSettings. F-keys via MainWindow::keyPressEvent (QML Shortcuts don't fire in the QQuickWidget docks). Commit 9f0d054.

Settings dialog

- Per-tab scroll-area + larger default size (long tabs were clipping).

Release

- Version bumped (CMakeLists.txt + installer/lyra.iss → 0.4.5), release note docs/releases/v0.4.5.md, User Guide CW section updated, annotated tag v0.4.5, installer dist/Lyra-Setup-0.4.5.exe, pushed to GitHub; main fast-forwarded to the release.

2026-06-21 — v0.4.3 RELEASED: 60 m band plan + tune-drive modes + BCD polish

Operator quality-of-life batch + release. All on main (project commits releases directly to main, tags + GitHub release + installer). Released mid-afternoon, then held for the day.

60 m band plan (`bandplan.{h,cpp}`, `PanadapterPanel.qml`)

- **Clickable 60 m channels:** US CH1–CH5 now render as always-visible

labelled bars (a touch larger), each a `MouseArea` that QSYs (`Prefs.mode= "USB" + Stream.setRx1FreqHz(qsy)`) to the channel's USB dial — same one-click idiom as the FT8/FT4/PSK watering-hole markers. `segments()` flags `channel/qsy` per sub-band; delegate gained `chan prop` + conditional label size/visibility + tooltip.

- **Region/country-correct 60 m:** precise US channel dial freqs

(5330.5/5346.5/5357.0/5371.5/5403.5 kHz); WRC-15 contiguous band (5351.5–5366.5) added to `r1Bands()` (was missing). New **Country** override layer: `regionBands(region) + countryBands(country) + bandsFor(region, country)` merge (replaces same-named band in place). `BandPlan::country()` + `bandPlanCountryChanged→regionChanged`; all 3 `bandsFor` call sites updated. UK = its 11 RSGB/Ofcom segments; Canada = WRC-15. `Prefs.bandPlanCountry` (key `band_plan/country`, default "AUTO", validated vs {AUTO,UK,CA}). Settings → Hardware Country combo added.

- Verify-don't-guess applied: UK 11-segment list `WebFetch`-confirmed

(`rs.gb.org`); `AskUserQuestion` before writing international freqs.

Tune-drive selector — #95 (`prefs`, `main.cpp`, `Profile`, **`settings`**, `TxPanel`)

- Replaced bool `useTuneDrive` with enum `tuneDriveMode`

{`Slider=0/Tune=1/Fixed=2`} + `fixedTuneDrivePct` (default 25), one-time `QSettings` migration (legacy bool → 0/1). `main.cpp` tune orchestrator rewritten to a `tunePctFor()` lambda + mode-aware arm/release (fixes a latent mode-switch-mid-tune restore edge). Profile capture/apply + `TxPanel` visible: `Prefs.tuneDriveMode === 1` updated.

- **Max-drive cap is honored:** wire chokepoint `setTxDriveLevel`

(`std::min(maxDriveRaw(), ...)`) already covers TUN; added cosmetic defense-in-depth — Tune/Fixed spinners `setRange(0, maxDrivePct())` + live follow on `maxDrivePctChanged`. So TUN can't exceed the operator's Max-TX-drive ceiling.

11 m → 10 m BCD option (`usb_bcd.{h,cpp}`, **`settingsdialog`**)

- New `elevenAsTen` (key `hw/bcd11as10`, default ON), mirroring the

existing `sixtyAsForty`. Substitution lives in `reapply()`: an 11 m/CB freq (`cbBandIndexForFreq≥0`, amateur table returns -1) remaps to the 10 m amateur index → emits **BCD 9**. Settings → Hardware checkbox.

- Plus an amber regulatory advisory under the BCD options: "Operate only

within the maximum power and band limits permitted by your country / region's regulations." (Operator-requested — states it plainly at the point of use re: 11 m abuse.) `USER_GUIDE` BCD section mirrors it.

Docs + release

- `USER_GUIDE`: Band-plan (Region+Country+60 m channels), Settings → TX

(Tune-drive subsection + Max-drive cap note), USB-BCD (11 m option + advisory). New `docs/releases/v0.4.3.md` (= GitHub release body).

- Version 0.4.2 → 0.4.3 (`CMakeLists.txt` + `installer/lyra.iss`); full

rebuild (LYRA_VERSION embedded, `USER_GUIDE` resource recompiled) + ISCC → `dist/Lyra-Setup-0.4.3.exe` (68 MB).

- 06921f2 Release v0.4.3 (16 files, +598/–158) + the already-on-main

help-doc commit 4e62bb9 (log-capture section) rode out together. Tag v0.4.3, pushed main (ddd6677..06921f2) + tag, GitHub release published with installer attached + verified.

Standing rules in effect

- NO PATCHING (tune-mode done as a proper enum + migration, not a shim).
- Commit/push only when asked — operator explicitly requested this release.

- Verify-don't-guess for regulatory data (60 m freqs, UK segments).

2026-06-19 — v0.4.1 RELEASED: TX protection (SWR cut + drive cap) + PHROT

Short session: shipped the TX-protection arc, a PHROT parity feature with a digital auto-gate, doc updates, then cut + pushed **v0.4.1** (main FF'd, tag + GitHub release + installer). All on main.

TX protection — #169 SWR + #170 drive cap (SHIPPED + operator-bench-confirmed)

- **#169 SWR Cut** (6c3e8e7): a 50 ms QTimer (swrEvalTimer_) armed on the

MOX edge in HL2Stream decodes `prn->tx[0].fwd/rev_power` → $p=\sqrt{(\text{rev}/\text{fwd})}$, $\text{SWR}=(1+p)/(1-p)$ (calibration-free). Trips through the single `requestMox(false)` unkey funnel above a user threshold (default 5:1). Four false-trigger guards: blank window, fwd/rev power floors (NaN-safe), dwell. Latch + reason + **PROT lamp** on TxPanel (gray/green/red, elided so it can't overflow the ATT lamp). Settings → TX group (enable + limit + Cut/Fold + fold-floor + during-tune).

- **#169 Phase 1b Fold**: optional action — steps drive $\times 0.5$ via

`applyDriveLevelNoPersist` (no QSettings write, fold restored next keydown) and escalates to a hard trip at the floor.

- **#170a drive cap**: `maxDrivePct` (1..100%) clamps every drive write at the

one chokepoint `setTxDriveLevel(std::min(maxDriveRaw(), ...))` — covers the operator slider, TUN, BandMemory restore, TCI DRIVE alike, + re-clamps live on lowering. Operator bench-confirmed ("ran slider max down, couldn't pull more"). **#170b over-power trip DEFERRED** by operator ("fine as it is").

- Design doc docs/architecture/tx_protection_design.md stamped COMPLETE.

PHROT — #109 phase rotator toggle + digital auto-off (SHIPPED)

- WDSP TXA `SetTXAPHROTRun` exposed via the wire/wdspcalls X-macro seam +

a `TxControl.setPhrotRun` callback (d99794d). `phrotEnabled` Q_PROPERTY (Settings → TX checkbox, default ON = WDSP/reference posture). Parity with the reference Setup PHROT control — it is fully exposed there, not a hidden knob (operator asked; verified).

- **Auto-off in digital** (165e176): one chokepoint `applyPhrotRun()`

computes `run = phrotEnabled_ && !(txMode==DIGU|DIGL)` (WDSP mode 7/9) and pushes via the callback — called from the operator setter, **every** `setTxMode` **edge**, and `registerTxControl` (channel open). The checkbox is now the operator's *voice-mode* intent; DIGU/DIGL switch it off on the wire and back on in voice modes (mirrors the native-rack `SetTxRackBypass` gate and the RX EQ #59 mode gate).

- **USER_GUIDE**: new Phase Rotator section incl. a **PHROT** and wide / ESSB

audio (4–10 kHz)*** note (helps punch on comms-grade wide SSB; all-pass group delay smears hi-fi ESSB → many ops leave it off; A/B on own voice).

Release

- v0.4.0 already shipped, so this is **v0.4.1**. Bumped CMakeLists +

installer.iss, rebuilt clean, ISCC → `dist/Lyra-Setup-0.4.1.exe` (68.1 MB).

- Commits 6c3e8e7 (SWR/drive-cap/#172/docs) → d99794d (PHROT toggle) →

165e176 (PHROT digital auto-off) → eff9a5d (version bump). Pushed main (1c6de15..eff9a5d), tag v0.4.1, GitHub release w/ installer.

- One snag: LNK1104 (operator had Lyra running) — asked operator to close, relinked clean (the standing don't-force-kill-operator-Lyra rule).

2026-06-18 — CW TX ON AIR: paddle + keyboard console + CW-over-TCI (#105)

CW transmit went from "design mapped" to fully on-air this session.

CW-2 paddle (gateway iambic) — SHIPPED + operator-confirmed

- `applyCwKeyerEnable` arms `prn->cw.cw_enable` in CW mode (CWL/CWU) — the

firmware-keyer + sidetone master; paddle in the HL2 KEY jack keys the carrier autonomously (gateway CWTX), host stays out of timing.

- Carrier-on-marker fix: keyed carrier rides the marker in both sidebands

(single shared `cwPitchHz`); the off-marker was a DISPLAY bug (host MOX off the gateway key line switched to the TX analyzer) — fixed by NOT forwarding host MOX in firmware-keyer CW (QSK). Break-in selector QSK(default)/Semi/Manual; CW MON sidetone slider on the Audio panel.

- Manual break-in foot-switch confirmed ("tone footswitch and paddle"):

foot switch = PTT hold (gateway PTTTX), paddle keys within (→CWTX).

- Vestigial `cwSidetoneFreqHz` path removed (14be1ed) — CW pitch is the single freq source.

CW-3a/3b host software keyer (CWX) + CW console — SHIPPED 8904749

- Gateway-verified (`radio.v` / `dsopenhpsdr1.v`): host CWX keys the carrier

via the SAME CWTX path as the paddle — host drives `tx[0].cw` (per element) + `cw_ptt` (held per message); `cw_enable` (CW-mode armed) enables BOTH the host EP2 overlay AND the gateway CWX decode (`cmd_data[24]` = the 0x0f C1 bit). NO WDSP / no host carrier gen. The earlier "dot/dash bit" guess was WRONG — tracing corrected it.

- `tx/CwMorse` (table + PARIS timing + operator weight, pure/testable),

`tx/CwKeyer` (dedicated-thread element pump, monotonic absolute-deadline scheduling, interruptible abort), `HL2Stream` `sendCw/abortCw` + `setCwxKey/setCwxPtt` + lazy keyer + abort-on-close.

- `qml/CwConsolePanel` — chip→floating console (audio-rack idiom): WPM,

`type+Enter` sends / `Esc` aborts, `Send+Stop`, reserved CW-5 decoder pane. "CW" launcher chip after the TX DSP strip.

- Design doc `docs/architecture/cw3_software_keyer_design.md`.

CW-4 CW-over-TCI — SHIPPED fcfcb6b, **EESDR-spec-verified**

- `tci_server` dispatch routes the EESDR TCI CW commands into the CW-3

keyer: `cw_macros` (plain send), `cw_msg` (contest prefix/call/suffix + \$N repeat), `cw_macros_speed/cw_keyer_speed` (WPM, bidirectional), `cw_macros_stop`. Reserved-char un-escape `^~*→:; ;`. `sendCw/abortCw` marshalled to the stream thread (`QueuedConnection`) — no race vs the console. CW-mode-only.

- **Verified vs the EESDR TCI protocol manual** (operator added)

`docs/TCI Protocol.pdf` §3.2) + `Thetis TCIServer.cs` + `SDRLogger+`. The EESDR cross-check caught 3 drafting errors: `cw_macros_empty` is a server→client terminal signal (not a stop), reserved-char un-escape was missing, and `cw_msg` (the canonical contest cmd) was missing.

Docs

- `docs/help/USER_GUIDE.md` (2a09824): new "CW operating" section

(paddle/console/TCI + break-in + CW MON) + "CW keying over TCI" + TOC.

Standing directive (recorded)

- TCI work verifies against BOTH `Thetis TCIServer.cs` AND the EESDR TCI manual (`docs/TCI Protocol.pdf`) — the authoritative spec.

Next

- **CW-5 (#173)** — RX CW decoder (faithful port of `SDRLogger+`'s

Bayesian/AFC/Farnsworth decoder, `hamlog/cw.html`) into the console's reserved pane + a macros field. Deep-read the decoder first.

- CW follow-ons: Semi/Manual host-MOX for CWX, F-key memories, `CWFWKeyer`

toggle; TCI `cw_terminal/cw_macros_delay/|..|+<>` macro syntax.

- On-air TCI CW bench via SDRLogger+ still TODO (spec-verified + builds).

Commit summary (today)

2a09824 docs(cw): USER_GUIDE — CW operating + CW-over-TCI
fcfcb6b feat(cw): CW-4 — CW keying over TCI, verified vs EESDR TCI spec
8904749 feat(cw): CW-3a/3b — host software keyer (CWx) + CW console
14be1ed refactor(cw): remove the vestigial `cwSidetoneFreqHz` path

2026-06-17 EOD — v0.4.0 RELEASED + main consolidated + CW TX mapped

UI readability batch (shipped 8fe2100)

- TX DSP launcher chips (Speech/EQ/Combinator/Plating) → boxed buttons that fill with the accent when their panel is open (were flat text-only).
- EQ graph: frequency-marker row across the top; x-range tracks the TX audio passband (SSB/digital = high edge, AM/DSB/FM/SAM = half occupied) + amber dashed TX-edge marker; gain scale bigger (bold 11px) + brighter, labelled every 3 dB (added $\pm 3/\pm 9$ rows), 0 dB bright blue.
- Speech/Combinator/Plate parameter labels + readouts brighter + bold 13px; Combinator per-band micro-labels 9/10→11px. Operator: "easier on the old eyes." `.gitignore += _run_lyra_vac1.bat`, Line.

main consolidated (operator request)

- origin/main fast-forwarded a3a517f..8fe2100 (clean, 32 commits, no foreign divergence). Local main moved up + checked out — main **is the working trunk again** (TX largely done; stops the "forgot to FF main" risk). `tx-rebuild` kept as a legacy ref.

v0.4.0 RELEASED

- Version bump 0.3.1→0.4.0 (CMakeLists + installer/lyra.iss); release notes `docs/releases/v0.4.0.md`; commit 88f17f2. Full rebuild (version baked), Inno Setup installer `dist/Lyra-Setup-0.4.0.exe` (~68 MB). Annotated tag `v0.4.0`, pushed main + tag, GitHub Release published with installer attached + verified. `main==origin/main==v0.4.0==88f17f2`.

#156 (restart-after-hard-kill) — PARKED

- Operator empirical: a real accidental hard kill of `lyra.exe` mid-VOICE-TX (my pre-build force-kill fired while they were tuning) → **restarted CLEAN**. Only ever one observed hang (2026-06-13 mid-TUN). Intermittent → parked until it reproduces; prime suspect noted (infinite `hReadThreadInitSem.acquire()`), optional bounded-timeout hardening if revisited. PROCESS RULE logged: don't force-kill `lyra` for a build when the operator may be on air — ask first.

CW TX (#105) — DESIGN MAPPED (rainy-night research, no code)

- 2-lane reference dive (Thetis host `cxw.cs/console.cs/TXA.c` + HL2 wire `networkproto1.c` + operator's HL2+ gateway RTL). Wrote `docs/architecture/cw_tx_design.md` — full port plan.
- THE finding: HL2 CW is **100 % gateway-keyed** (no host/WDSP carrier; `TXA.c` `SetTXAMode` has no CWL/CWU case; `radio.v:1145` bypasses EP2 TX-I/Q during CW, feeds the FPGA envelope ramp; FPGA also makes the hardware sidetone). Host just sends key-state. **The verbatim EP2 CW packer is ALREADY in tree** (`NetworkProto1.cpp:99-108`, dormant) + `prn->cw` seeded; EP6 dot/ptt decode already avoids the reference's left-shift bug. Missing: CW C&C; composer cases (`0x0f/0x10/0x0b`) + the native CWX keyer engine + FSM CW PttSource + UX. Build order C-1..C-5 in the doc. ■ flagged: `0x0b` register collision with the shipped ATT-on-TX step-att — resolve before composing.

2026-06-17 (cont. — AM/DSB/FM native TX modulation + AM carrier #106/#93)

#165 follow-on → *basic WDSP-native AM / DSB / FM transmit*

Operator: keying AM showed only the upper sideband. RX was correct; the one-sided signal was the operator's own TX — the modulator was SSB-only.

- `hl2_stream.cpp` `setTxMode`: widened the WDSP-mode clamp `0..1` → `0..13` so AM(6)/DSB(2)/FM(5)/SAM(10) reach `setTXAMode` instead of being pinned to USB.
- `main.cpp` `wdspTxModeFor`: full mode map (LSB0...SAM10, default USB), replacing the AM/FM/DSB/SAM→USB collapse.
- `main.cpp` `pushTxFilter`: per-mode sign-coded bandpass (TX mirror of the RX §14.2 convention) — USB-side `+low..+high`, LSB-side `-high..-low`, double-sideband symmetric. NEVER calls `SetTXABandpassRun` (§15.23 trap); `SetTXABandpassFreqs` + `SetTXAMode` configure `bp0`. FM CTCSS silenced via `SetTXACTCSSRun(ch, 0)` (TXA defaults it ON at 100 Hz).
- **TUN** ■ `cw_pitch_offset` generalized in `txDdsHzForTune / txAnalyzerOffsetHz` so double-sideband modes stay centred.
- New wire bindings: `SetTXACTCSSRun`, `SetTXAAMCarrierLevel` (`wdspcalls.{h,cpp}` — pointer + X-table + extern).

Doubled-bandwidth bug (operator bench) — FIXED

AM straddled centre correctly but occupied ~2× the set TX BW. Cause: the symmetric branch used $\pm\text{high} = 2 \times \text{high}$ total. `txf->high` is the set TX BW (`Prefs.txBandwidth`); a double-sideband signal occupies the full $\pm\text{span}$, so the edge must be $\pm\text{high}/2$ — mirroring `WdspEngine::computePassband(half = bw/2,  $\pm\text{half}$ )` which also draws the filter markers. Now AM at 6k = $\pm 3\text{k}$ = 6k total, inside the markers. Operator bench: **correct**.

AM carrier level control (#93) — operator-facing, reference-faithful

- `HL2Stream` `amCarrierPct` `Q_PROPERTY` (+ getter/setter/signal/member, `QSettings tx/amCarrierPct`), `TxControl.setAmCarrierLevel` callback, `main.cpp` lambda → `SetTXAAMCarrierLevel`. Settings → TX "AM Carrier" spinbox (0–100 %).
- **Mapping matches the reference exactly** (verified at `setup.cs:9846`): UI is % of **standard carrier POWER**, coefficient = $\sqrt{(\text{pct}/100)} \times 0.5$, **default 100 = standard AM** (= 25 % of PEP). So operator-stored values transfer 1:1. Helper `amPctToCarrierLevel()`, applied at both push sites. (Earlier draft passed `pct→coeff` straight, default 50 — corrected before bench.)
- N8SDR's exported profiles confirm the intent: 8K-AM-N8SDR=40,
*-N8SDR-AM=84, AM 10k CFC=95 (reduced/controlled carrier).

Profile import — decided NOT to pursue (operator call)

DSP chain is Lyra-native (EQ/Speech/Combinator/Plate, not WDSP/CFC), so importing reference profiles is pointless — the `*sound*` lives in the native rack, which has no reference equivalent. `amCarrierPct` NOT added to the Profile struct. Profile manager stays as-is.

Status

All built clean + operator-bench-confirmed (AM/DSB/FM both sidebands inside the markers; AM carrier behaves). Committed this session; #106/#93 closed.

2026-06-17 (cont. — doc reconciliation + ATT-on-TX UI §15.31)

Doc reconciliation (operator-delegated while out)

- `THETIS_DIRECT_PORT_PLAN.md` reconciled (agent pass + my accuracy)

corrections): obbuffs/P1 → [DONE]; compress/cfcomp/eqp → [N/A] (native-superseded); wcpagc leveler/ALC → [DONE] but **softened** — metering "shipped in code; #160 bench close-out + Brent/Timmy field-confirm still IN PROGRESS" (not over-claimed). PDF + DOCX regenerated via the generalized `tools/sync_execution_plan.py` (now `--md/--docx/--pdf-path/--title/--no-docx`). `SESSION_LOG.pdf` regenerated.

§15.31 — ATT-on-TX operator surface + visible lamp (operator-flagged)

Operator: "ATT on TX doesn't appear to be live — I don't see the LNA drop to -31 on TX." TRACED the path (did NOT re-assert from the doc — the [DONE] I'd let stand was operator-disproven):

- **WIRE path IS live + correct:** `fsmAdvance` raises

`setTxStepAttnDb(31)` on keydown (`hl2_stream.cpp:1651`) → `tx_step_attn=0` → XmitBit-gated `compose_case_11 C4 = 0x40` → `ak4951v4 cmd_addr 0x0a rx_gain = 0 = min LNA during TX`. The front end WAS being attenuated.

- **Root of the report = VISIBILITY gap** (operator confirmed "1 and 3"):

the LNA readout shows the RX setpoint (unchanged on TX); no on-screen confirmation, where Thetis visibly jumps the ATT to 31. Also the `src/tx/AttOnTxPolicy.{h,cpp}` class is **unused / wire-inert** — the live mechanism is inline in `hl2_stream.cpp`; the doc citation was wrong.

- **FIX** (operator-locked design confirmed before code; matches Thetis

Setup → General → Ant/Filters → "ATT on Tx" ✓ / "ATT: 31"):

- `HL2Stream: attOnTxEnabled_` (default ON) + `attOnTxDb_` (default

31), `Q_PROPERTY` + setters + `QSettings(tx/attOnTx*)`; `fsmAdvance` gates on the toggle + uses the value (replaces hardcoded `kAttOnTxDb`); mid-TX toggle re-applies live. Disabled → axis 0 (= reference `SetTxAttenData(0)`).

- `TxPanel.qml`: "ATT" lamp in the gap between the Mic and Tune sliders,

AUTO/TUN/MOX lit idiom — gray ATT off / orange ATT 31 armed (RX) / solid-red + white-text ATT -31 engaged (keyed). Engaged colour fixed from a salmon-on-maroon blend that read PINK → MOX-style solid red; label shows the negative (applied attenuation) on TX per operator request.

- `settingsdialog.cpp`: Settings → TX → "ATT on TX (RX-ADC protection)"

group (left/safety column) — Enable + ATT dB (0..31) spin, bidirectional with the lamp.

- Builds clean; operator-approved the UI in review. THETIS row kept

[WIP] (flips [DONE] only on the on-air bench confirm). `AttOnTxPolicy` class still unused (flagged for later delete-or-wire-up). Task #114 (broader: panadapter offset + PA-enable safety) stays pending.

2026-06-17 (build day — #162/#163 ship, #90 TX monitor 3-route, #164 Out picker; all HL2-confirmed)

Branch `tx-rebuild` (NOT pushed; `main/origin` stays `v0.3.1`). Every commit below was operator-bench-confirmed on the HL2+ before it landed.

#162 TX Speech profile NOT saving — FIXED (root cause CORRECTED vs the 06-16 theory)

The 06-16 "binding-break / UI-refresh" theory was **WRONG**. Operator pushback ("are we even capturing the Speech panel?") + a registry dump (`HKCU\Software\N8SDR\Lyra-cpp\profiles\item\`) proved it: the speech blob saved all-default. **Real root cause = QML signal shadowing** — the Stage card's custom `signal toggled(bool)` was shadowed by `AbstractButton.toggled`, so the ON button's `onClicked: toggled(check)` emitted the Button's signal, never the model write → Speech always saved defaults.

- `c17960c`: qualify the `emit stage.toggled(check)` → the model write

fires; registry-verified (`deessOn:true ... now persist`).

- `46a428f`: `Binding{target:en; property:"checked"; value:on}` re-asserts

the toggle lamp from the model on recall (a checkable Button flips its own checked on click, severing the inline binding).

- Operator-confirmed: toggles save, recall re-lights lamps + restores

sliders, 2 fresh profiles round-tripped. Lesson: clean build + `qmlint` ≠ proof — verify by registry dump / launch.

#163 Rack-lamp dim in digital/CW + EQ analyzer recolor (cosmetic, operator-requested)

- 9be1fc1: Speech/Combinator/Plate panels gray the ON lamp + dim controls

+ show an amber "bypassed (MODE)" header hint in DIGU/DIGL **and CW** (digital → SetTxRackBypass; CW → no mic). Purely visual off a WdspEngine.mode binding; model untouched, re-lights on USB/LSB.

- 558d5d9: EqPanel — same dim + the operator's analyzer recolor:

Accumulate peak-hold → RED, pre-EQ "Before" overlay → WHITE (were near-white/amber), live "After" stays cyan. Line + RTA. Doc 66adc5e.

#90 TX audio monitor — ALL THREE reference-faithful routes SHIPPED (#90 closed)

ONE shared post-rack tap (xcmaster case 1, READ-ONLY on pcm->in, after the rack / before fexchange0) → lock-free SPSC MonitorRing (src/dsp/MonitorRing.h); drained ONCE at the top of dispatchAudioFrame into monScratch_, shared by all routes. WdspEngine monEnabled_ / monVolume_ (atomics, persisted); Audio-panel MON TX toggle + Monitor slider.

- Route 1 (HL2 jack, 6e6a282): monitor mono on L/R in place of the

auto-muted RX when MOX up + MON on, via the existing OutBound(0) path.

- Routes 2+3 (e1fd186): Route 2 = VAC stream-2 + SetIVACmon/

SetIVACmonVol (ivacGet-gated, re-applied in rebuildVac1); Route 3 = TCI RX-audio (tap moved into dispatchAudioFrame; MON→monitor / TX→mute / RX→live).

- **Reference-VERIFIED (operator asked "is this Thetis or have we drifted?"):**

Thetis audio.cs:386-404 drives SetIVACmon(0,1)+SetIVACmonVol(0, monitor_volume) on the PRIMARY VAC (id 0 = our kVac1Id) and SetTCIRxAudioMox/Mon (368-404) — the single MON drives jack + VAC + TCI, exactly as built. NOT drift.

- ■ Faithful behavior change: TCI RX-audio is now MUTED on the air (was

live RX during TX). Operator-bench-confirmed harmless — MSHV FT8-over-TCI decoded + completed a QSO (N8SDR↔KB6DAD to 73) with MON TX on, TX 5.1 W, DIGU rack auto-bypassed. Docs 266d8cd.

#164 "Out" output-device picker wired + MON→MON TX relabel (740f769, doc dc45485)

The disabled placeholder "Out" button is now a live one-click output picker: a styled top-level popup (popupType: Popup.Window so it ESCAPES the short Audio-panel QQuickWidget clip — a plain Item-type popup was getting chopped at the panel bottom with no scroll). Lists HL2 jack + PC devices, current ■/cyan/bold, hover highlight, as-needed scrollbar. Reads the same audioOutputDevices()/setAudioOutputDevice/audioDeviceIndex the Settings → Audio combo uses; full config stays in Settings. MON button relabeled **MON TX** so its purpose is clear. ComboBox considered + rejected (long device names blow out the row width).

New items logged (not started)

- **#165 (■ bug):** AM/DSB modes only produce the UPPER sideband (lower

missing). AM/DSB are double-sideband → the WDSP passband for those modes is likely set one-sided (USB-only) instead of symmetric -W..+W. RX-DSP investigation, queued next.

- **GitHub / main branch:** operator has questions to raise on return.

Docs

- USER_GUIDE updated for the rack dim/recolor, the monitor (MON TX + the

VAC/TCI routes), and the Out picker.

- This session-log entry + a THETIS_DIRECT_PORT_PLAN.md reconciliation

pass (mark done-but-unmarked items, add a post-P4 shipped-work entry). .pdf/.docx of both NOT regenerated (no pandoc/doc-gen tool on this box) — MD sources updated; PDF/docx regen pending the operator's tool.

2026-06-16 (design + investigation day — NO commits; two items parked for tomorrow)

Context: native TX DSP rack (EQ/Speech/Combinator/Plate) + #49 Profile Manager bundle in flight; operator field-testing profiles on HL2+.

#90 TX audio monitor — DESIGN LOCKED, parked for tomorrow's build

Operator ask: MON button + Monitor volume slider on the Audio panel upper row. Studied Thetis MON / MonitorVolume (gates processed TX audio into the RX audio mixer + SetIVACmon; the verbatim `xMixAudio(0,0,...,out[2])` "mix monitor audio" line is already carried at `CMaster.cpp:458`, fed empty `out[2]`). Operator chose **both: HL2 jack now, separate PC device later**.

- **Key finding:** `WdspEngine::dispatchAudioFrame()` (`wdsp_engine.cpp:3100`)

STAYS LIVE during TX — RX audio is just gain-zeroed by the `txMuted_ && autoMuteOnTx_` gate, then `OutBound(0) → EP2 LR → AK4951` jack. So the jack is live-but-silent during TX = the inject seam (no fight with RX stand-down).

- **Locked design — ONE shared tap:** in `xcmaster` case 1 capture

`pcm->in[stream]` AFTER the rack, BEFORE `fexchange0` (the processed "what you sound like" mic audio) → small SPSC ring.

- Route 1 (HL2 jack, NOW): drain ring in `dispatchAudioFrame`; when MOX

up + MON on, output monitor audio (scaled by new Monitor-vol) in place of muted RX audio → existing `OutBound(0)` path. No new stream.

- Route 2 (separate PC device, LATER): feed same tap into

`xvacOUT(kVac1Id, stream 2)` (replace `vacMonSilence_`) + `SetIVACmon = IVAC Stage 5 / DL-4` done reference-faithfully.

- Tradeoff (note in tooltip): post-rack tap is pre-WDSP-ALC/bandpass —

hear your rack DSP faithfully, not WDSP's corrective ALC.

- Operator: **"hold for tomorrow's build."**

#162 Speech-panel profile not saving — ROOT-CAUSED, parked first-thing tomorrow

Operator: TX Speech panel options aren't saved/recalled by Profiles while EQ/Combinator/Plate are. Decisive repro: turned De-esser ON, recalled an all-off profile → **De-esser button stayed ON**.

- Traced the full path: capture (`main.cpp:752`) / apply (`main.cpp:798`) /

`Profile toJson+fromJson+sameValues / SpeechModel saveState+loadState / single instance / context property` — **all symmetric & correct**. The data DOES save and DOES apply to the model + DSP.

- **Root cause = QML binding-break (UI-refresh, not persistence):**

`SpeechPanel` (and `PlatePanel/CombinatorPanel`) use `Button{checkable:true; checked:} + Slider{value:}`. The first click/drag makes Qt Quick write `checked/value` imperatively, **severing the binding to the model**. Recall then updates `model+DSP+numeric` labels but NOT the toggle/slider visuals — the button is a dead light. `EqPanel` dodges this (non-checkable toggles + pure `checked: binding off a rev counter + Q_INVOKABLE` getters). De-esser is the clearest demonstrator because a stuck binary toggle is unambiguous; `Plate/Combinator` share the latent pattern but only surface after mid-session interact-then-recall.

- **Fix (#162, high pri):** adopt `EqPanel`'s binding-safe pattern in the

three rack panels. QML-only, no DSP/wire risk.

Tomorrow's order (operator-set)

1. **#162** Speech (+Plate/Combinator) panel binding-safe rebuild — FIRST.

2. **#90** TX monitor build (Route 1 HL2 jack; Route 2 PC device as the

follow-on within #90 / IVAC Stage 5).

Memory: `project-lyra-cpp-monitor` (the #90 locked design).

2026-06-13 (Saturday — ■ TX BRING-UP COMPLETE + v0.2.3 RELEASED)

`main == tx-rebuild == HEAD e8aafbc (pushed)`. Tags: milestone `tx-working-2026-06-13` (at `bd61d07`) + release `v0.2.3` (at `a5dc763`). GitHub Release **v0.2.3 — "Transmit: SSB + digital (TCI) on the air"** PUBLISHED with `Lyra-Setup-0.2.3.exe` (67.7 MB) attached, marked Latest.

P4.b Wire-LIVE switchover — SHIPPED + operator HL2 bench PASSED

- fb9ec41 P4.b-1 — Wire-LIVE RX-out gate (operator bench PASSED).

- 0b551b4 P4.b-2 — TX wire-LIVE: first modulated RF through the direct-port chain (operator HL2 bench).

- 9db2c50 P4.b-2.1 — TX SSB sideband fix: USB transmits USB (9100-verified).

- 84dbb12 fix(tx): TUN panadapter spike on-marker — TX-analyzer crop

shifted by NCO-dial offset. Verified Lyra TUN is **Thetis-exact zero-beat** (two-offset scheme; the bug was display-only — authoritative refs: console.cs:32553-32587 UpdateTXDDSFreq path, postgen 30788-30800, readout 22035-22051).

- §7 dead-code retirements (3 build-gated commits): fe6a438

OutboundRing + Ep2SendThread, 99fd24a txWorkerLoop + composeCC + buildEp2KeepaliveTemplate, e711256 old HL2-audio-jack EP2 ring (pushAudio + setHL2AudioSink).

- P4.b bench gates PASSED: SSB voice both sidebands + Phase-3-EXIT

RF-safety kill-test (RFG drops on hard-kill).

TCI digital TX re-home — SHIPPED + ON-AIR VALIDATED

- bd61d07 — filled the §10.3 src/tci/TciTxBridge.{h,cpp} skeleton

(singleton, mono deque, l=Q=mono drain w/ zero-fill underrun, queuedSamples() for CHRONO). Registered via SendpInboundTCITxAudio(&TciTxBridge::inboundCb) after create_cmaster; fed by tci_server.cpp handleBinaryFrame tap; gated by SetTXTCIAudio(0, micSource=="tci"). The verbatim DSP seam (xcmaster case 1 use_tci_audio gate) was already present — only the host-side sink was missing.

- **Bench: MSHV → TCI → FT8 real on-air QSOs** — N3YDN (FM29),

F4FLF (JN18, ~4160 mi DX), KO4OIG (EL89), W3EWL/QRP.

Space-bar PTT toggle (#157) — SHIPPED

- 2824ebf — Prefs.spaceBarPttEnabled (tx/space_bar_ptt_enabled,

default ON), MainWindow keydown/keyup gate, Settings → Hardware → Transmit checkbox under the HW-PTT tickbox. Operator-verified working.

Release

- e8aafbc — version bump 0.2.2 → 0.2.3 (CMakeLists project(VERSION) +

installer/lyra.iss AppVersion). Clean build, ISCC installer, tag, GitHub release (notes + installer).

NEXT (morning)

- **Mic-input paths discussion** (PC/VAC host audio + plain-HL2 vs HL2+).

Key reframe: a **plain HL2 (no AK4951) has no radio-side mic at all** → host audio-in (#102 VAC1) is REQUIRED for those ops to TX phone, not optional. All sources are interchangeable 48 k mono feeders of the same InboundTCITxAudio seam (TciTxBridge = the template). §5.4 "single mic-source class, no abstraction" rule has expired now that there are 4 concretes — introduce ITxAudioSource. Capability-aware default (plain HL2 → PC/VAC; HL2+ → AK4951 mic). Decide: TX-in-only vs full bidirectional VAC; Qt Multimedia vs WDSP rmatchV.

- Parked: #156 restart-after-hard-kill-mid-TX hang; intermittent

menu-drag display stutter (monitoring; ruled out as a §7 regression);

#104 HL2 Line In (codec mux); #103 VAC2.

2026-06-12 (Friday PM-4 — P4.a prep SHIPPED, both commits wire-inert)

Branch: tx-rebuild, HEAD baef866 (pushed).

P4.a-1 SHIPPED a4aa8c3 — *sendProtocol1Samples verbatim (DORMANT)*

- NEW src/wire/NetworkProto1.cpp — networkproto1.c PARTIAL: the EP2 writer thread sendProtocol1Samples verbatim (networkproto1.c:1204-1267). Mechanical diff IDENTICAL (whitespace-normalized, accommodations mapped back); sole structural delta = braces around the DEFERRED non-HL2 else-branch (comment-only branch needs braces to compile). Accommodations documented in the preamble: lyra::wire namespace; the HPSDRModel enum-class mapping; write_main_loop_hl2(prn->outBufp) = the FrameComposer monolithic WriteMainLoop_HL2 equivalent (#121/#122 fold); generic WriteMainLoop (non-HL2/ANAN P1) carried as DEFERRED reference text; (AVRT_PRIORITY) cast + suppress 4100 on the verbatim signature.
- io_keep_running global added (network.h:411 verbatim; init 0 so the loop body cannot run even if started prematurely) + the decl in RadioNet.h.
- FrameComposer write_main_loop_hl2 tail flipped from the Step-14 outbound_notify_consumed_pair() translation to the verbatim ReleaseSemaphore(prn->hobbufsRun[0/1], 1, 0) pair. Dormant-safe: nothing calls write_main_loop_hl2 until P4.b starts the writer; the hobbufsRun handles are created at the P4.b open() with the quartet.
- DORMANT: nothing _beginthreadex's the function until P4.b's open() (reference start site = StartAudio, netInterface.c:66 + the quartet :68-71). Build clean, zero new warnings.

P4.a-2 SHIPPED baef866 — *PostGen seam entries + verbatim field types*

- **wdspcalls**: SetTXAPostGen{Run,Mode,ToneMag,ToneFreq,TTMag,TTFreq} for the P4.b TUN re-home (legacy DC-injection TUN dies with the legacy EP2 packer; TUN becomes the reference TXA output-side tone generator). Pre-cdef audit: NO header declares these — PORT-exported definition sites in wdsp/gen.c only (gen.c:784/792/800/808/817/826), signatures harvested from the definition bodies; presence + exact casing verified against the bundled wdsp.dll export table (PE scan, all 6 PRESENT of 533). TT pair included per table rule #3 (PS committed; the PS calibration drive is the two-tone).
- **TxReadBufp**: std::vector -> verbatim double* (network.h:62) + the verbatim calloc at create_rnet (netInterface.c:1604, 2*sizeof(double)*720, process lifetime — the P2.b OutBufp precedent). Ep6RecvThread's 5 .data() sites -> bare pointer; the vector-size guard reduces to a null check (fixed 1440-double capacity >= 4*spr for every valid nddc).
- **hWriteThreadMain**: std::thread -> verbatim HANDLE (network.h:90). Zero users existed on the std::thread form; P4.b assigns the verbatim _beginthreadex(sendProtocol1Samples) handle.
- Build clean (sole warning = the pre-existing hl2_stream C4456, untouched file). Commit-message nit: a backtick-quoted literal got eaten by bash substitution again (the P2.b class) — citation survives, no force-push.

NEXT = P4.b (THE switchover — ONE commit, full operator HL2 bench gate)

- open(): quartet CreateSemaphore x4 + prn->hWriteThreadMain = _beginthreadex(sendProtocol1Samples) (io_keep_running=1 BEFORE start).
- dispatchAudioFrame -> the asioOUT-pattern tee (HL2-jack = real audio into OutBound(0); PC = zeroed tee so EP2 pacing never starves) — the B.6.b-delicate RX switchover.
- Inbound(inid(1,0), mic_sample_count, prn->TxReadBufp) at the live EP6 mic-harvest site (Ep6RecvThread, the block already mirrors networkproto1.c:560-579).
- FSM re-home: TXA SetChannelState arming per the \$15.25 ground truth + TUN -> SetTXAPostGen.
- Control-plane mapping per the design doc \$5 table; retirements per \$7 (txWorkerLoop + keepalive + DC-TUN + slew-fill, OutboundRing, Ep2SendThread, FrameComposer cv tail).
- Teardown: io_keep_running=0 -> release both hsend sems once -> join -> CloseHandle quartet -> destroy_obbufs BEFORE destroy_xmtr.
- Bench gate per design doc \$8 incl. explicit TX-state lines + FIRST VOICE through the chain.

2026-06-12 (Friday PM-3 — P2 audit complete + P2.a eer completion SHIPPED)

Branch: tx-rebuild, on top of 2897756.

P2 fidelity audit (the deliverable that scopes P2.b/c/P4)

- **Reference TX-out chain (HL2 P1), fully read:** ob_main -> sendOutbound (network.c:1237; P1 branch :1285-1340) — id 1 memcpys 126 complex into prn->outIQbufp + Release(hsendIQSem) + Wait(hobbufsRun[0]); id 0 -> outLRbufp + hsendLRSem

+ hobbufsRun[1] -> sendProtocol1Samples thread (networkproto1.c:1204-1267): WaitForMultipleObjects(both, TRUE) -> eer overwrite if peer->run && XmitBit -> !XmitBit => memset outIQbufp -> swap_audio_channels -> 16-bit BE quantize + HL2 CW-bit overlay into OutBufp -> WriteMainLoop_HL2 -> MetisWriteFrame -> Release both hobbufsRun. Resources: buf alloc'd netInterface.c:1606-1608; semaphore quartet created in StartAudio :68-71; obbufs rings 0/1 created in UpdateRadioProtocolSampleSize :1856-1857; eer created run=0 per xmtr (cmaster.c:212-224).

- **Lyra dormant Step-14 surface:** functionally-parallel idiom translations predating the verbatim mandate — prn buffers are std::vector (RadioNet.h:552-554), the semaphore quartet became OutboundRing's bool+cv, Ep2SendThread is the sendProtocol1Samples rewrite. XmitBit is already verbatim (extern int, RadioNet.h:733). Disposition: re-port verbatim, retire the translations at P4 Wire-LIVE.
- **Landmine found:** BOTH reference functions deref pcm->xmtr[0].peer->run, but P0.d had deferred the eer creation — peer was NULL. That made P2.a unambiguous and first.

P2.a SHIPPED — eer completion (verbatim)

- wire/cmcomm.h: opaque eer typedef replaced by the FULL verbatim struct (wdsp/eer.h:30-49, mechanical diff IDENTICAL); opaque DELAY twin typedef added (wdsp/delay.h:32-59).
- wire/wdspcalls.{h,cpp}: +5 seam entries — create_eer / destroy_eer / xeer / pSetEERSize / pSetEERSamplerate (signatures verbatim from eer.h:51-75; all five verified present in the bundled wdsp.dll via PE export-table scan).
- wire/CMaster.cpp: all four deferred eer sites RESTORED verbatim — create_xmtr create_eer(run=0, ...) 12-arg block (cmaster.c:212-224); destroy_xmtr destroy_eer (:262, after destroy_ilv per reference order); xcmaster xeer(pcm->xmtr[tx].peer) (no-op at run=0 with in==out — verified in eer.c:86-122); SetXmtrChannelOutrate pSetEERSamplerate/pSetEERSize. Restored-lines verbatim-subset check PASS (18 lines).
- Runtime impact: create_xmtr now also creates/destroys one run=0 eer object per xmtr (heap + CS only); xeer in the TX pump body never executes today (stream-1 pump has no producer). RX path untouched. Clean build, zero warnings.

P2.b SHIPPED (same session) — prn outbound surface verbatim

- outLRbufp/outIQbufp/OutBufp std::vector -> the VERBATIM reference pointer fields (network.h:64-66) + the verbatim calloc set at create_rnet (netInterface.c:1606-1608, mechanical subset PASS). Lifetime = process (create_rnet call-once) = reference.
- Verbatim HANDLE quartet declared on prn (hsemLRsem/hsemIQsem/hsemEventHandles[2]/hobbufsRun[2], network.h:92-95) — DORMANT/nullptr until P4's StartAudio equivalent; the Step-14 cv+flags translation stays alongside until P4 retires its consumers.
- Callers bent: OutboundRing .data() *6 -> raw; Ep2SendThread pack casts -> char; write_main_loop_hl2 parameter -> const char* (reference type). Wire-inert (only create_rnet's allocation runs live). The lone C4456 in hl2_stream.cpp:776 is PRE-EXISTING (file untouched; surfaced by the header-triggered recompile) — flagged for a separate cleanup.

P2.c SHIPPED (same session) — sendOutbound verbatim + ob_main restore

- NEW src/wire/Network.cpp — network.c PARTIAL direct port: sendOutbound verbatim (network.c:1237-1341). P1 branch LIVE (the outLRbufp/outIQbufp memcpy + hsemLRsem/hsemIQsem release + hobbufsRun wait handshake, incl. the EER de-interleave sub-branch with its function-local static ptr); ETH branch carried as DEFERRED reference text (Protocol 2/ANAN = v0.4; WriteUDPFrame/udpOUT unported). Accommodations, each documented inline: the RadioNet.h enum-class mapping on the protocol gate; suppress 4101 (ETH-only locals) + 4456 (the reference's own loop-scope i shadowing its function-scope i, network.c:1239 vs :1298). Mechanical diff vs network.c:1237-1341 = IDENTICAL (whitespace-normalized, deferred text restored).
- wire/ObBufs.cpp ob_main: the DEFERRED sendOutbound(id, a->out) hand-off RESTORED verbatim — the P1 TU is now reference-complete.
- Still dormant end-to-end: no create_obbufs caller until P3/P4; the semaphore quartet is created by P4's StartAudio equivalent before the pump can deliver, the same ordering the reference relies on. Clean build, zero warnings.

P3 SHIPPED (same session; survived + re-verified after an operator power outage mid-arc) — netInterface registrations + obbufs ring lifecycle

- **create_rnet moved to once-per-process** at the main.cpp QTimer block AFTER create_xmtr — the reference's C#-init ordering (create_cmaster -> create_xmtr -> create_rnet -> StartAudio). The old per-open call re-allocated prn on EVERY stop/start (leak + wire-state reset, contradicting the close() "prn stays alive for re-open" contract — a latent pre-existing defect the P3 reference-read surfaced). open() now asserts the reference prn-non-null contract (netInterface.c:40); if wdsp.dll failed to load, open() refuses with a qCritical (no WDSP = no radio, the reference posture).

- **SendpOutboundTx(OutBound) restored at the reference site** (netInterface.c:1761, tail of create_rnet) — pcm->OutboundTx -> xmtr[0].pilv via the verbatim no-guard SetILVOutputPointer; ordering safe by construction. The Stage C.3-fix "registration belongs at the xmtr-open site" note SUPERSEDED (it guarded a Step-14 stub against per-open clobber; under the direct port the reference site is correct). RX side deliberately NOT registered — WdspEngine's openRx1 owns RX dispatch in the approved hybrid until P4 (registering RX here = the B.6.b clobber = dead RX audio).
- **UpdateRadioProtocolSampleSize ported verbatim** (netInterface.c:1836-1858, mechanical diff IDENTICAL with the documented kMax*/enum-class accommodations) — called per session-open from HL2Stream::open() at the StartAudio position (:45): per-protocol spp values (USB: mic 63 / rx 63 / tx 126 / audio 126) + create_obbuffs(0/1) — TWO new ob_main pump threads per session, both idle (ring 0 has no producer until P4's RX switchover; ring 1's producer is xilv in the stream-1 cm pump, quiescent until P4 feeds Inbound(1,...)). destroy_obbuffs(0/1) at close() per StopAudio (:112-113).
- Power-outage recheck: all 4 edited files probe-verified intact, diff-vs-HEAD = exactly the P3 hunks, forced rebuild of the touched TUs clean (sole warning = the pre-existing hl2_stream C4456), mechanical diffs re-PASSED post-outage.
- [DONE] **OPERATOR HL2 BENCH PASSED (same day):** "rx fine" — RX working on the P3 lifecycle (once-per-process create_rnet, per-session obbuffs rings + 2 idle pump threads, TX registration live). **P3 gate cleared; P4 Wire-LIVE unblocked.**

■ Operator-flagged near-miss + bench-discipline correction (2026-06-12)

- Operator accidentally pressed TUN a few sessions back and got REAL RF OUT, despite the rebuild-arc bench notes repeatedly saying "TX stays wire-quiescent / RX-only." Code-verified: those statements were true of the NEW direct-port chain only. The OLD hardware-validated TX-0c path — TUN DC-injection at the EP2 packer (hl2_stream.cpp:~2516) + the requestMox FSM + ATT-on-TX — was NEVER part of the TX rip (only the SSB voice DSP subsystem was removed) and has been LIVE on tx-rebuild the whole time. RF required the operator's own persisted PA-enable opt-in + drive settings (the deliberate-arm gate worked as designed; it was armed from the first-RF benches and persisted).
- **Operator decision: LEAVE IT** — it is the proven path, gated behind the PA-enable opt-in; P4 replaces it under its own full bench. (Disarm anytime: un-tick Enable PA in Settings.)
- **Discipline correction going forward:** every direct-port bench gate now includes an explicit "TX state unchanged" line (TUN/MOX behavior + PA-enable posture), and "RX-only / wire-quiescent" is reserved for statements about the WHOLE wire, not just the new chain. Applies to the P4 gate and onward.

P2 COMPLETE. NEXT

1. **P3** — netInterface outbound registrations: SendpOutboundRx(OutBound) / SendpOutboundTx(OutBound) per netInterface.c:1749-1761 — MUST register AFTER create_xmtr (verbatim setters have NO null guards); plus the create_obbuffs(0/1) sites (netInterface.c:1856-1857) at the Lyra equivalent of UpdateRadioProtocolSampleSize.
2. **P4** — Wire-LIVE switchover (one commit, full HL2 bench gate): verbatim sendProtocol1Samples thread + the semaphore-quartet creation + WriteMainLoop_HL2 hand-off; retires the OutboundRing/Ep2SendThread cv translations.

2026-06-12 (Friday PM — P1 obbuffs.c verbatim direct port SHIPPED)

Branch: tx-rebuild, on top of 976062d (the P0.d bench-PASS doc flip).

Shipped (one commit)

- **NEW** src/wire/ObBuffers.{h,cpp} — reference obbuffs.{h,c} verbatim: obb, *OBB twin typedef + numRings/obMAXSIZE/OBB_MULT defines + the file-scope _obpointers obp four-alias bank + create/destroy/flush/OutBound/obdata/ob_main/SetOBRingOutsize. The TX-OUT seam (WDSP output → ring → ob_main pump thread → sendOutbound → protocol packers); SEPARATE TU from cmbuffs (inbound-only).
- The reference's own **2014-era idioms kept as shipped** (deliberately NOT harmonized to the cmbuffs sibling): calloc/free (not malloc0/_aligned_free), destroy-time obp.pcbuff[0]==NULL guard, obdata WITHOUT the MWOLGE CS wrap, the csOUT enter/leave pair at the top of the ob_main loop, the out work buffer inside the struct.
- **Sole deferred line:** sendOutbound(id, a->out); in ob_main — the reference defines it at network.c:1237 (the P1 branch :1285-1340 is the outIQbufp/outLRbufp + hsendIQSem/hsendLRSem/hobbuffsRun handshake WriteMainLoop_HL2 consumes) = **P2 scope**. Carried in place as reference text with a DEFERRED tag; decl in ObBuffers.h verbatim.
- Provenance check resolved before code: the "OutBound at network.c:1285-1340" citation in older RadioNet.cpp comments is the **PROTOCOL_1 branch of sendOutbound**, not a second OutBound — obbuffs.c's OutBound is the one registered via SendpOutboundRx/Tx (P3).

- Compiler accommodations: same set as CmBufs.cpp (suppress 4312/4189/4311/4302 on the thread-arg casts + (AVRT_PRIORITY) 2), each marked inline.

Verification

- Clean build, ZERO warnings.
- Mechanical diff vs reference: **ObBufs.cpp IDENTICAL** (code lines, whitespace-normalized); ObBufs.h sole delta = the reference's include-guard `#define _obbufs_h`, replaced by `#pragma once` (documented packaging difference, same as CmBufs.h). Live-line-subset PASS on both.

Wire impact

- **NONE — the TU is dormant.** No Lyra caller of `create_obbufs` exists until P3/P4 (the reference creates rings 0/1 in `netInterface.c:1856-1857`). No new threads at startup, wire bytes unchanged. An operator RX smoke at convenience is prudent (binary changed) but this touches no live path.

NEXT

1. **P2 = sendOutbound / sendProtocol1Samples fidelity audit** of the dormant wire layer (reconcile `network.c:1237-1341` + `networkproto1.c sendProtocol1Samples` vs Lyra's Step-14-era `OutboundRing/MetisFrame/Ep2SendThread` surface), then restore the `ob_main` hand-off verbatim.
2. P3 `netInterface` registration (`SendpOutboundRx/Tx(OutBound)`) — AFTER `create_xmtr`, the verbatim setters have NO null guards) → P4 Wire-LIVE (one commit, full HL2 bench gate).

2026-06-12 (Friday — P0.d CmBufs/CMaster/cmsetup verbatim direct port SHIPPED + ■ HL2 RX-regression bench PASSED)

Branch: `tx-rebuild HEAD = afc7950`, pushed to `origin/tx-rebuild`.

■ **OPERATOR HL2 BENCH (same day): RX-regression gate PASSED** ("RX still working") — clean RX on the new per-stream `cmbufs` pump/ring layout (two `cm_main` threads, stream 0 idle by design). P0.d is fully closed; **P1 (obbufs.c verbatim port) is UNBLOCKED** and starts next session.

Shipped (one commit, `afc7950`)

- **NEW** `src/wire/cmsetup.{h,cpp}` — reference `cmsetup.{h,c}` verbatim: `cmMAX*` sizing macros (16/4/4/2/32), `rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize`, `SetRadioStructure` + `set_cmdefault_rates`. `CreateRadio/DestroyRadio` carried with the unported `pipe/sync` calls commented.
- `src/wire/CmBufs.{h,cpp}` **rewritten verbatim** — `cmb`, `*CMB` twin typedefs, `#define CMB_MULT (3)`, `malloc0/_aligned_free`, no `calloc/intptr_t/guard` deviations; `pcm->in[]` allocation moved back to `create_cmast` (reference shape).
- `src/wire/CMaster.{h,cpp}` **rewritten verbatim** — FULL `_cmast` struct (`cmaster.h:39-99`, PS surface incl. `out[3]/panalloc/pgain/peer`), `cmaster`, `*CMaster`, raw TCI fn ptrs, enum `AudioCODEC`, `cm = {0}`. **TxChannel RAIL carve-out DELETED** (`src/wdsp/TxChannel.{h,cpp}` retired): verbatim no-arg `create_xmtr` opens the WDSP TXA channel (`chid(1,0)=1`) + TX analyzer (`disp 1`; pan analyzer = `disp 0`, no collision) + `out[0..2]` + `create_ilv` itself through the `wdspcalls` seam. `create_cmast/destroy_cmast` verbatim per-stream loops (`update[] CS + cmbufs + in[]` for all `cmSTREAM` streams; `create_rcvr` = deferred stub, RX hybrid). `xcmaster` verbatim: `update[]` critical section restored, real `stype/txid/chid`, TCI-override memset restored; `fexchange0` + monitor-mix `xMixAudio` (accept-gated, quiescent vs the 1-input `WdspEngine` mixer) + `xilv` live. `SetXcmInrate/SetCMAudioOtrate/SetRcvr-/SetXmtrChannelOtrate/SetRunPanadapter/SetAntiVOXSource*` ported. All deferred subsystem lines carried IN PLACE as reference text with DEFERRED tags.
- `src/wire/cmcomm.h` — opaque verbatim twin typedefs `ANB/NOB/EER/VOX/TXGAIN/ANALYZERS` (tags match reference headers; completing a type later is source-compatible) + the umbrella-mapping note (`.cpp` files include explicit family headers — an include-list umbrella would be circular under `#pragma once` because the family headers include `cmcomm.h` for the base surface).
- **main.cpp** — `SetRadioStructure(2,1,1,1,0,...)/set_cmdefault_rates(48 k)` config block BEFORE `create_cmast` (derived ids match the live layout: `chid(0,0)=0` = `WdspEngine` RX1 channel, `chid(1,0)=1` = TX channel); `create_xmtr()` invoked in the `QTimer` block after `resolve_wdsp_calls` (the documented DEFERRED-CALLSITE accommodation); `handler-1.5` = gated `destroy_xmtr()`. Consumer-side decls for the headerless PORT functions (`test_ilv` precedent). `RadioNet.cpp` → enum `AudioCODEC` cases; `scratch/test_ilv.cpp` → verbatim `cmast cm = {0}`; globals.

Verification

- Clean build, ZERO warnings (C4701 in getChannelOutputRate disabled function-wide with a documented verbatim-text-wins pragma — line-level suppress can't reach the code-gen-stage warning).
- scratch/test_ilv.exe ALL PASS against the new verbatim globals.
- Mechanical diff vs reference: CmBuffs.{h,cpp} / cmsetup.h / CMaster.h struct+decls+enum / all 10 fully-verbatim cmaster.c functions IDENTICAL (whitespace-normalized); cmsetup.cpp comment-only deltas; live-line-subset check PASS for the partially-deferred bodies. Sole code accommodation: (char *)"" at the XCreateAnalyzer call (C++ string-literal constness).

Behavior changes at startup (for the bench)

- TWO cm_main pump threads now start (streams 0 + 1; stream 0 idles forever — no Inbound producer; the reference layout). Stream rings sized from cmMAXInRate=384000 (in[] = 512 complex).
- TX stays wire-quiescent: no Inbound() producer yet, pcm->OutboundTx null until P3.
- destroy ordering: handler-1.5 destroy_xmtr() (gated) → handler-4 destroy_cmater() = reference relative order.

NEXT

1. ~~Operator HL2 RX-regression bench (the P0.d gate)~~ — ■ **PASSED 2026-06-12** (RX working; gate cleared).
2. **P1 = obbuffs.c port** (TX-out seam, separate TU from cmbuffs) → P2 sendOutbound audit → P3 netInterface registration (outbounds AFTER create_xmtr) → P4 Wire-LIVE (one commit, HL2 bench gate).

2026-06-08 (Monday EOD — STAGE B aamix.c port COMPLETE + bench-validated)

Branch: tx-rebuild HEAD = 533b06b. Pushed to origin/tx-rebuild (0/0 in sync). origin/main deliberately untouched (different arc). **Backup:** _backups/lyra-cpp-2026-06-08-aamix-stage-B-COMPLETE.bundle (16 MB, git bundle --all).

Today's shipped commits (in order, all on tx-rebuild)

```
533b06b [aamix port -- Stage B.6.b-final] strip diagnostic instrumentation
8b8e0da [aamix port -- Stage B.6.b-fix1] remove Stage-A NO-OP SendpOutboundRx stub (THE FIX)
12369bc [aamix port -- Stage B.6.b-debug-sink] dispatchAudioFrame sink-side counters
225518c [aamix port -- Stage B.6.b-debug] 9-counter chain instrumentation
53d9e5a [aamix port -- Stage B.6.b-retry] wire AAMix via reference-faithful path
6e773a5 Revert "[aamix port -- Stage B.6.b] wire AAMix(1-input passthrough) into RX path"
525c2f0 [aamix port -- Stage B.6.b] wire AAMix(1-input passthrough) -- REVERTED at bench
27ac2fa [aamix port -- Stage B.6.a] extract feedIq audio tail into dispatchAudioFrame
```

Arc summary

- **B.0 → B.5:** AAMix.h/cpp port complete from earlier sessions (538-line header verbatim from aamix.c with GPL v3+ attribution to NR0V/WDSP, 1100+ lines of cpp; idiom translations from Win32 to C++23: CRITICAL_SECTION→std::mutex, HANDLE semaphore→std::counting_semaphore, _beginthread→std::jthread, _Interlocked*→std::atomic with fetch_or/fetch_and, malloc0→std::vector RAII).
- **B.6.a (27ac2fa):** Refactored feedIq audio tail (lines 2402-2456) into new WdspEngine::dispatchAudioFrame(audio, nframes) helper. Byte-identical relocation; bench-confirmed.
- **B.6.b first attempt (525c2f0):** Wired aaMix_ into RX path with create_aamix(active=0x01L) shortcut. **Operator bench: NO audio.** Reverted (6e773a5).
- **B.6.b-retry (53d9e5a):** Reference-faithful per cmaster.c:297-313: create_aamix(active=0L, ring_size=4096, slew=0/10/0/10ms) → SetAAudioMixOutputPointer(aaMix_, 0, outbound) → SetAAudioMixState(aaMix_, 0, 0, 1) (activate via close_mixer/open_mixer slew atom). **Operator bench: STILL NO audio.**
- **B.6.b-debug (225518c):** Chain-side instrumentation — 9 atomic counters + 1 ENTRY beacon + producer/consumer 1-Hz log emits in xMixAudio / mix_main. **Operator bench result:** chain healthy end-to-end (xmix=rel=wake=out=188/sec, nzOut=99%, mixmain_started=1, Outbound=set) but operator confirmed STILL no audio. ⇒ defect is downstream of Outbound.
- **B.6.b-debug-sink (12369bc):** Dispatch-side instrumentation — 6 atomic counters in dispatchAudioFrame + 1-Hz log line tag [disp-dbg]. **Operator bench result:** [disp-dbg] lines NEVER fired in the entire 33-second capture. ⇒ dispatchAudioFrame is never entered. ⇒ the Outbound callable AAMix is calling is NOT the lambda WdspEngine wired.

• **B.6.b-fix1** (8b8e0da, **THE FIX**): Root cause traced from bench timestamps:

- 19:22:02.972 — WdspEngine::openRx1 calls create_aamix(id=0, ..., Outbound=dispatchAudioFrame_lambda) ⇒ paamix[0]→Outbound = real lambda ✓
 - 19:22:02.973 — HL2Stream::open() → create_rnet() → SendpOutboundRx(STUB) ⇒ pcm→OutboundRx = STUB, then SetAAudioMixOutputPointer(nullptr, 0, STUB) resolves nullptr → paamix[0] ⇒ aaMix_→Outbound = STUB, **real lambda CLOBBED X**
 - The Stage-A NO-OP stub at RadioNet.cpp:272-283 was a placeholder ("Stage B aamix port wires it via SetAAudioMixOutputPointer") for the wire-up Stage B was supposed to replace. Stage B.6.b correctly wired dispatchAudioFrame in openRx1 — but left the Stage-A stub registration in place, which then overwrote the wiring 1 ms later when HL2Stream::open() called create_rnet().
 - **Fix:** delete the Stage-A stub registration. Comment block on the deleted case warns future hands that the wire-up needs the per-channel sink-routing context (hl2Out_, hl2AudioPush_, audioRing_) that the global pcm cannot reach — so the wire-up must live at openRx1, not create_rnet.
- **B.6.b-final** (533b06b): Strip diagnostic scaffolding. -134 lines. Brief comment in AAMix.cpp points future hands at commit history (git show 225518c for chain instrument, git show 12369bc for dispatch instrument).

Operator HL2+ bench (final, 19:27:49 → 19:28:47 — 58-second run)

- HL2 jack startup: first [disp-dbg] post-unmute: calls=188 muted=0 gain=0.0845 peak16=32767 hl2(push/null)=188/0 hl2pushSet=1 audioRingSet=0 — full-scale audio reaching AK4951.
- Operator-confirmed audible RX ("Yes we have audio now").
- Mute toggle via dispatch (muted counter 0→22→133→287).
- Output device switched to PC Soundcard at 19:28:17 → hl2out=0, audioRingSet=1, pc(push) 176→928/sec.
- Switched back to HL2 jack at 19:28:23 → hl2out=1, hl2(push) resumed climbing.
- AGC mode changes (med/slow/off/fast) clean.
- Chain out=N tracks 1:1 with dispatch calls=N across both sinks across the entire run — no dropped frames, no thread starvation.

Tasks closed

- #130 [completed] Stage B.6 [DEFER] — migrate RX audio path to ported AAMix
- #132 [completed] Stage B.6.b — wire AAMix(1-input passthrough) into RX path
- Stages B.0–B.5 + B.6.a already closed in earlier sessions.

Methodology lessons recorded (for future Stage X work)

- 2-stage instrumentation (chain + dispatch) localized the defect to a single line of code on the SECOND bench. No speculation cycle, no convergence theatre.
- Operator-empirical rule held: after first B.6.b failure I was tempted to revert the AAMix port itself; the chain counters proved AAMix was correct in isolation, redirecting the dig to the wire-up surface where the bug actually was.
- Hard rule for next port: counters first when the symptom is "nothing happens". Instrumentation cost is trivial vs the alternative (multi-round revert/speculate cycle).
- "Do as Thetis does, Lyra-Native style" continues to hold — the first B.6.b attempt's active=0x01 shortcut cost a revert cycle; the reference's active=0 then SetAAudioMixState(activate) pattern worked first time on the retry.

Memory + doc updates

- MEMORY.md index line for lyra-cpp-tx rewritten with "READ FIRST 2026-06-09 AM" header + branch/HEAD/backup/root-cause pointer.
- project_lyra_cpp_tx.md — new "■ READ FIRST 2026-06-09 AM resume pointer" block at top, plus full "State (2026-06-08 EOD)" section. Earlier state preserved below.
- EXECUTION_PLAN.md — new "PROGRESS TRACKING — STAGE B (aamix.c port) [SIDE-TRACK]" section with full B.0 → B.6.b-final checklist; new row in VERIFICATION LOG; footer "Status as of 2026-06-08 EOD" updated.

- This SESSION_LOG entry.
- PDF/DOCX regenerated via tools/sync_execution_plan.py.

Next session pointer

- **First action:** `git fetch origin THEN git status + git log --oneline -5 tx-rebuild origin/main` to surface divergence.
- **Next port target = OPERATOR'S CALL.** Stage B side-track done; candidates:
 - **Stage C:** `ilv.c` port (TX I/Q interleaver — pairs with the `SendpOutboundTx` stub already wired in `CMaster.cpp` Stage A)
 - **Stage D:** `xcmaster` pump body port
 - Or pivot back to Step 14 / Phase 3 wire-layer rebuild on main (previous-arc resume was TX-1 Component 7 — first end-to-end SSB voice TX)

Wire status

RX audio flows live through the ported AAMix dispatcher on the `tx-rebuild` branch's `HL2Stream::open()` path. TX path unchanged from earlier `tx-rebuild` state.

2026-06-06 (Saturday LATE-AFTERNOON — §1-C FULLY COMPLETE + RE-AUDIT CLEAN)

Branch: `tx-rebuild` HEAD = `169f1c2`. Build clean, `lyra.exe` relinked at 11:10:09. **16 commits shipped today** + TWO full multi-agent audit rounds. §1-C arc COMPLETE through Stage 4F.2 (FrameComposer dissolution). Final re-audit returned 37/37 PASS across 3 sections, zero remaining §6-Q3/Q5-class candidates, zero forbidden tokens in `src/wire/`.

What's reference-faithful now

- TX wire layer: 100% reference-faithful (all `_radionet` fields in `RadioNet`; all reference file-scope globals as TU-scope statics in correct `.cpp`; Router + OutboundRing + FrameComposer all dissolved into free functions matching reference; symmetric socket-binding via `metis_wire_bind()` in both `Ep6RecvThread` + `Ep2SendThread`).
- §1.1 verdict: ■ → ■ PARITY (Stage 4E doc consolidation).

What's NOT yet reference-faithful (tracked, NOT in §1-C)

- **Task #73** — pre-existing forbidden-token violations
(Thetis / Console.cs / OpenHPSDR / PowerSDR) in 5 non-wire files: `tci_server.{h,cpp}`, `mainwindow.cpp`, `hl2_stream.{h,cpp}`, `wdsp_native.h`. Doc-style cleanup, ~1 hour, no impact on shipped wire-layer paths.
- **Wire-INERT** — nothing in `HL2Stream::open()` calls the new wire-layer free functions yet (step 14 wire-up).
- **Task #114 deferrals** — EER mode, non-HL2
`WriteMainLoop_generic`, `PeakFwdPower/PeakRevPower` helpers. Hardware-blocker (no ANAN hardware to bench-test).

Step 14 scope (NOT trivial)

`HL2Stream` is 4203 lines total with its own monolithic `rxWorkerLoop` + `txWorkerLoop` member methods at `hl2_stream.cpp:921 + :2437` (the OLD wire path). Step 14 must REPLACE these (per Rule 7 "Delete TX, don't refactor") with calls into the new wire-layer free functions:

1. `prn` global pointer assigned at session-open
2. `metis_wire_bind(socket, dest, dest_len)`
3. `outbound_init()` to size `prn->outLRbufp/outIQbufp`
4. `ForceCandC::prime(3, tx_freq, rx_freq)` synchronously
per §7 temporal-separation contract

5. `Ep6RecvThread::start(socket_fd)` → spins RX thread

reading from `prn-> + WSAEventSelect`

6. `Ep2SendThread::start(socket, dest, dest_len)` → spins

TX thread; no more composer / ring params (Stage 4F.2)

7. Migrate HL2Stream's existing telemetry/mic/IQ callbacks

to the new sink-based registration (`Router::register_sink` for IQ; `Ep6RecvThread::set_*_sink` for telemetry/mic/I2C)

Plan-first commit (R6). Bench-critical (first time the new wire layer actually emits datagrams to the radio). Multi-hour, possibly multi-commit. Wrong wire-up = broken RX/TX on real HL2 hardware.

Resume after step 14

Operator-bench verification of RX-side audio + telemetry on real HL2+ hardware. Then TX-side bench (CW/SSB + Palstar power readings). Then Task #73 cleanup (small, doc-only, can happen anytime).

Wire status

Wire-INERT. All wire-layer components reference-faithful but not connected to runtime.

2026-06-06 (Saturday AFTERNOON — §1.1 REVERT COMPLETE — historical)

Branch: `tx-rebuild HEAD = 79fe4ec`. Build clean, `lyra.exe` relinked at 10:45:10. **TWELVE commits shipped today** + 6-agent comprehensive TX audit done. ALL stages of the §1-C sweep complete in code (§1.1 networking- infrastructure exclusion fully reverted). Only Stage 4E (PARITY_CHECKPOINTS doc consolidation) remains before step 14 wire-up.

Today's commits (newest first)

- `79fe4ec` §1-C Stage 4D — `OutboundRing` dissolves to free functions (§1.1 revert COMPLETE)
- `1d73bea` §1-C Stage 4C — `Ep2SendThread` reads from `prn->OutBufp + g_fpga_write_bufp`
- `f1031da` §1-C Stage 4B.1 — correct `FPGAReadBufp` scope (file-scope, not `_radionet`)
- `26acea3` §1-C Stage 4B — `Ep6RecvThread` reads from `prn-> + WSAEventSelect`
- `e157301` §1-C Stage 4A — add networking-infrastructure fields to `RadioNet`
- `d88ddd6` §1-C Stage 3 — `OutboundRing` wait-all via `condition_variable`
- `2b78d9e` §1-C Stage 2A — `Router` class → struct + free functions
- `0e2a375` §1-C Stage 1 — small reverts (`wb_enable`, push timeout, diag counters)
- `50b713b` §6-B nit — drop `metis_write_frame` null-guard
- `3bfd248` §6-B + §7 — TU-scope `MetisFrame` + `ForceCandC` populate

§1.1 revert summary

All `_radionet` fields the original §1.1 exclusion (signed ■ 2026-06-04) had punted to wire-layer class members now live in `RadioNet` as members exactly per reference:

- Buffer pointers: `RxBuff`, `TxReadBufp`, `outLRbufp`,
`outIQbufp`, `OutBufp` (+ `ReadBufp` declared P2-future-use)
- RX seq counters: `cc_seq_no`, `cc_seq_err` (P2-only;
declared in `RadioNet` but HL2-inert)
- Thread handles: `hReadThreadMain`, `hWriteThreadMain`,
`hKeepAliveThread` (std::thread types)
- Init semaphores: `hReadThreadInitSem`, `hWriteThreadInitSem`
(std::counting_semaphore<1>)

- Outbound sync (cv-collapsed mirror of 4 reference HANDLE

pairs): cv_outbound + mu_outbound + 4 bool flags + outbound_stop

- WSA event + waitable timer (Win32-only): hDataEvent, wsaProcessEvents, hTimer, liDueTime

- Networking config ports: p2_custom_port_base, base_outbound_port

Reference-FILE-SCOPE globals (NOT in _radionet) live as TU-scope statics in their respective wire-layer .cpp files per the §6-B precedent:

- wire/MetisFrame.cpp: g_metis_out_seq_num (was Ep2SendThread::out_seq_num_)
- wire/Ep6RecvThread.cpp: g_metis_last_recv_seq, g_seq_error, g_seq_seen, g_control_bytes_in[5], g_fpga_read_bufp
- wire/Ep2SendThread.cpp: g_fpga_write_bufp

Wire-layer classes dissolved or reduced:

- Router: class → plain struct + free functions (Stage 2A)
- OutboundRing: class → namespace-scope free functions

operating on prn->... (Stage 4D)

- Ep6RecvThread: still a class (thread orchestrator), but

all data moved to prn->... or TU-scope statics

- Ep2SendThread: same — thread orchestrator, data moved out

Remaining — Stage 4E (doc-only)

Consolidate the §1-C entry in docs/architecture/PARITY_CHECKPOINTS.md:

- Single §1-C entry covering Stages 1+2A+3+4A-D + the earlier §6-B + §7
- §1.1 verdict amended from ■ OPERATOR-APPROVED DEVIATION to ■ PARITY (now genuinely byte/structure-faithful)
- Sign-off block + commit trailer

Resume after Stage 4E

Phase 2 step 14 — wire outbound_init() + metis_wire_bind() + ForceCandC::prime + Ep6RecvThread::start + Ep2SendThread::start into HL2Stream::session_open() in the correct temporal sequence.

Wire status

Wire-INERT today. All wire-layer components built but nothing in HL2Stream::session_open calls them yet.

2026-06-06 (Saturday MID-DAY — STAGE 3 CHECKPOINT — historical)

Branch: tx-rebuild HEAD = d88ddd6. Build clean, lyra.exe relinked at 09:54:09. Five commits shipped today + 6-agent comprehensive TX audit done. Stages 1+2A+3 of the §1-C sweep DONE; Stages 2B+4 remain (bundled — §5.10 WSAEventSelect lives in _radionet fields per reference, so it lands with §1.1 revert).

Today's commits (newest first)

- d88ddd6 §1-C Stage 3 — OutboundRing wait-all via condition_variable
- 2b78d9e §1-C Stage 2A — Router class → struct + free functions
- 0e2a375 §1-C Stage 1 — small reverts (wb_enable, push timeout, diag counters)
- 50b713b §6-B nit — drop metis_write_frame null-guard

- 3bfd248 §6-B + §7 — TU-scope MetisFrame + ForceCandC populate

What got fixed to match reference (Stages 1+2A+3)

- §1.3: `std::atomic wb_enable` → `volatile long` (matches `network.h:160`)
- §6.10: OutboundRing producer push — drop bounded 5s `try_acquire_for` + `push_timeouts_*`; unbounded `acquire()` mirroring reference `Inbound/obuffs`
- §6.12: `Ep2SendThread` `datagrams_sent/send_errors` diagnostic counters dropped (no reference counterpart)
- §5.9: `class Router` → plain struct + free functions
(`xrouter/register_sink/set_control_word/set_call_count/router_instance`) + file-scope `g_routers[]` array, mirroring reference `router.c` structure verbatim
- §6.9/6.11/6.14: OutboundRing wait-all via `std::condition_variable` + bool flags + single mutex (was 4 `binary_semaphores` + polling) — mirrors reference `waitForMultipleObjects(2, hsendEventHandles, TRUE, INFINITE)` atomic wait-all, no polling, fully interruptible

Remaining — Stage 4 (THE BIG ONE)

§1.1 networking-infrastructure revert + §5.10 WSAEventSelect bundled.

§1.1 (signed ■ 2026-06-04) excludes from RadioNet a list of fields that the reference puts INSIDE `_radionet`. Under strict "do as reference, period" §1.1 itself is a §6-Q3/Q5- class candidate. Stage 4 puts these fields back into RadioNet:

- Buffer pointers: `RxBuff`, `TxReadBufp`, `ReadBufp`, `OutBufp`, `outLRbufp`, `outIQbufp`
- Thread/sem/event handles: `hReadThreadMain`, `hsendEventHandles[2]`, `hobbuffsRun[2]`, etc.
- Waitable timer + WSA event: `liDueTime`, `hDataEvent`, `wsaProcessEvents`
- RX seq counters: `cc_seq_no`, `cc_seq_err`
- Networking ports: `p2_custom_port_base`, `base_outbound_port`

Then `Ep6RecvThread` + `Ep2SendThread` + OutboundRing read from `prn->...` instead of their own members. §5.10 (`recv()` → `WSAEventSelect`) folds in because `hDataEvent` + `wsaProcessEvents` are `_radionet` fields.

Scope: multi-hour, multi-commit. Planned sub-stages:

- 4A: Add fields to RadioNet (additive, no behavior change)
- 4B: `Ep6RecvThread` refactor + `WSAEventSelect` mechanism
- 4C: `Ep2SendThread` refactor (reads buffers from RadioNet)
- 4D: OutboundRing — buffers + sync primitives move into

RadioNet; class may dissolve into free functions

- 4E: PARITY_CHECKPOINTS §1-C entry finalized + §1.1 amendment

Resume after Stage 4 completes

Phase 2 step 14 — wire `ForceCandC::prime` + `metis_wire_bind` + `WSAEventSelect` setup into `HL2Stream::session_open()` in the correct temporal sequence.

Wire status

Wire-INERT today. All wire-layer components built but nothing in `HL2Stream::session_open` calls them yet.

2026-06-06 (Saturday MID-DAY — AUDIT PAUSE BOOKMARK — historical)

Branch: `tx-rebuild HEAD = 50b713b`. Build clean, `lyra.exe` relinked at 09:27:22. All §6-B + §7 work shipped + audited 20/20 PASS this morning.

Done this morning (2 commits, ALL like reference)

- 3bfd248 §6-B parity correction sweep + §7 ForceCandC populate

- Hoisted `metis_write_frame` to `wire/MetisFrame.{h, cpp}` TU-scope
- TU-scope `g_medis_out_seq_num` (was `Ep2SendThread::out_seq_num_`)
- TU-scope `socket/dest` globals (were `Ep2SendThread` members)
- DROPPED `Ep2SendThread::send_lock_mutex` (no reference counterpart)
- §1.1 row split: RX seq stays / TX seq moves
- `ForceCandC::prime + prime_pass` byte-for-byte from `networkproto1.c:106-139`
- 50b713b §6-B nit — dropped `metis_write_frame` null-guard (no reference counterpart; audit-caught NOTE-D)
- Two parallel audits run: parity (20/10 PASS) + rule-compliance (10/10 PASS)

Pause reason

Operator-requested **comprehensive TX audit** of ALL work since the rebuild started (30 commits since 0348238 Phase 0 mapping doc 2026-06-04). Sections to audit against the reference:

- §1 RadioNet (2171fb9/dcbf9d1)
- §2 RbpFilter / RbpFilter2 (228ffd8)
- §3 HPSDR family + dispatch globals (f3ccc51/55ae658/782e65d)
- §4a/§4b-1/§4b-2/§4c FrameComposer 19-case dispatch (93e1511/5fc4192/4446916/f06a796)
- §5 / §5-A Ep6RecvThread + Router (c8baa63/a6be425/b988177/6bbf449)
- §6 / §6-A Ep2SendThread + OutboundRing (4c580a1/89ab298)
- §6-B + §7 already audited this morning (skipping re-audit)

Resume point after audit

Resume here: **Phase 2 step 14 — wire-up** `ForceCandC::prime + metis_wire_bind` **into** `HL2Stream::session_open()` in the correct temporal sequence (bind → prime → Ep6 start → Ep2 start).

Wire status

Wire-INERT today. Nothing in `HL2Stream::session_open` calls the new wire-layer components yet. Step 14 (next commit after audit closes) flips the wire on.

2026-06-05 (Friday EOD)

Project: lyra-cpp TX Wire-Layer Rebuild (Task #112) — branch `tx-rebuild`

Done today

- **§5 EP6 Receive Path** — `Ep6RecvThread + Router` (`xrouter/twist`) populated (c8baa63)
- **§5-A 9-fix parity correction** (a6be425) including ■ CRITICAL RX-audio bug (IQ unpack scale 48 dB too quiet — caught before any consumer wired)
- **§5-A doc completeness** (b988177)
- **§6 EP2 Send Path** — `Ep2SendThread + OutboundRing` populated (4c580a1)
- **§6-A 6-fix parity correction** (89ab298) — MMCSS priority, per-family dispatch, CW bit-shift revert to verbatim, CW state per-sample reads, `binary_semaphore` UB guard, Rule 24 inline token
- 4 audit passes total (2 per §5 / §6) — all PASS after corrections

Where we are

Branch tx-rebuild HEAD = 89ab298. §5 + §6 of the wire-layer rebuild are **shipped + doubly-verified** (parity-audit + rule-audit + correction sweeps + re-audits, all clean). Wire-layer is still wire-inert — no Phase 1 code path instantiates the new components.

Next up

§7 — ForceCandC priming + ANY remaining wire-layer pieces. Reference: `networkproto1.c::ForceCandCFrame` (lines 134-139) + `ForceCandCFrames` (lines 106-132). Small commit — just the 3-frame priming burst that runs at session start before the EP2 send loop begins.

After §7 completes the §1c-listed wire-layer skeletons, the next phase is **Phase 2 step 14 — wire-up**: instantiate `Ep6RecvThread` + `Ep2SendThread` + `OutboundRing` + `Router` + `ForceCandC` + `FrameComposer` into `HL2Stream`'s session-open path.

Pending / parked

- **Task #114** TX-policy plumbing — ATT-on-TX, panadapter offset, PA-enable safety, plus the §6 FIXMEs: `writeMainLoop_generic` for non-HL2, EER mode (§6.4), `PeakFwdPower/PeakRevPower` helpers (§5 telemetry cases 0x08/0x10)
- **No hardware bench yet** — gateway-watchdog kill-test, real-antenna TX, foot-switch HW-PTT all parked until Phase 2 wire-up + verification
- **Don't push to GitHub** (standing rule)

Commit summary (today)

```
89ab298 fix(wire): §6-A parity correction sweep – 6 fixes
4c580a1 feat(wire): populate §6 Ep2SendThread + OutboundRing
b988177 docs(parity): §5-A Lyra-native additions table – 2 audit-flagged rows
a6be425 fix(wire): §5-A parity correction sweep – 9 fixes
c8baa63 feat(wire): populate §5 Ep6RecvThread + Router
```

Standing rules in effect

- Rule 24 "always verify against reference" — applied per commit, audit-verified
- 2026-06-05 directive "reference = make Lyra the same" — driving all Q-decisions
- §6 EER mode + non-HL2 dispatch + Peak power helpers all deferred to Task #114 with FIXMEs
- Operator hardware = HL2+/AK4951; no ANAN P1/P2 hardware available

Session log started 2026-06-05. Older entries (if any) below this line as they accumulate.